

A Hybrid Deep Learning Model for Image Dehazing and Object Detection to Enhance Autonomous Vehicle Navigation

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Abstract—Autonomous vehicle navigation demands precise environmental perception, which is especially challenging due to the presence of fog. To address the deterioration of visual quality due to adverse weather, a hybrid deep learning approach is presented that combines Non-Local Image Dehazing (NLD) and the state-of-the-art best object detection algorithm YOLOv10. The NLD method first improves visibility from fog images using non-local patch similarity, which restores scene details effectively. These dehazed images are then fed into the YOLOv10 model for later precise object detection. Conventional models can fail in low visibility cases, leading to reduced detection accuracy and compromised navigation decisions. The proposed approach avoids these limitations by dehazing the images, thus supporting more robust feature extraction and detection. Comparative evaluations indicate that integrating NLD significantly improves detection performance under foggy conditions. This hybrid model is a significant step towards the creation of safer and more trustworthy autonomous driving systems, especially in real-world setups interrupted by weather-related disturbances.

Keywords— Non-Local Image Dehazing, YOLOv10, Object Detection, Autonomous Vehicle Navigation, Foggy Weather Conditions.

I. INTRODUCTION

Autonomous vehicle navigation has advanced significantly in the recent past primarily because of advancements in computer vision and deep learning algorithmic solutions. These technologies rely primarily on high-quality visual data for the provision of object detection and robust decision-making. However, real-world scenarios often include adverse weather conditions such as fog, which severely diminishes visibility and impairs the effectiveness and safety of the navigation systems. Not only does fog reduce contrast and visibility of visual information, but also reduces the performance of conventional object detection algorithms, thus making reliable perception a Significant challenge. To address these limitations, image dehazing has emerged as a crucial technique in vision-based applications. Non-Local

Image Dehazing (NLD) is one efficient method in this area that uses non-local patch similarity to restore scene details that are disturbed by atmospheric interference. When combined with state-of-the-art object detection algorithms, such dehazing methods can significantly improve detection accuracy in adverse environmental conditions.

A hybrid deep learning approach is proposed, which integrates the Non-Local Image Dehazing (NLD) method with the latest YOLOv10 object detection model. The NLD module enhances the visibility and reconstructs key scene structures from hazy input images and is a key component of the overall model. The enhanced representations are fed into YOLOv10 to enable efficient and real-time object detection. The combined architecture surpasses the weaknesses of conventional models under low visibility conditions by enhancing feature extraction and detection accuracy and robustness.

The proposed system was first trained on a non-foggy dataset to develop a reliable object detection foundation under clear visibility conditions. Subsequently, the model was validated using foggy datasets, where the Non-Local Image Dehazing (NLD) module played a vital role to recover scene clarity prior to detection. The training and testing process assisted in rigorous examination of the combined framework in low-visibility conditions. Comparative performance against baseline models shows that incorporating the NLD approach greatly enhances detection accuracy and reliability in foggy weather, thus enhancing system robustness in real-world autonomous driving scenarios. Through enhanced environment perception under severe weather, the model results in more reliable and safer autonomous driving.

A. Objectives

- To integrate the Non-Local Image Dehazing (NLD) method into YOLOv10 to improve object detection in fog conditions.

- To train the YOLOv10 model on non-foggy datasets and test its performance using dehazed foggy datasets to access real-world feasibility.
- To improve detection performance, accuracy, and reliability in low visibility conditions by taking advantage of dehazing as well as the synergistic benefit of deep-learning-based detection.

II. LITERATURE REVIEW

Haze-affected object detection has been explored using classical and deep learning techniques. Berman et al. [1] introduced learning-free Non-Local Image Dehazing (NLD) using color clustering in haze-free images but not real-time. Zhang et al. [2] introduced NLDN, using non-local modules in CNNs but with high complexity. Wang et al. [3] introduced YOLOv10 for real-time detection with multi-scale optimization but without tuning the hazy condition. Zhang et al. [4] introduced IDOD-YOLOv7, a two-stage pipeline that enhances detection in fog but raises inference time. Rath et al. [5] introduced ODD-Net using CNNs and attention mechanisms but based on synthetic haze data. Khan et al. [6] proposed SD-YOLO-AWDNet for dehazing and detection simultaneously, with good performance in fog but not severe weather. Li et al. [7] proposed YOLO-Vehicle-Pro with multimodal input and cloud-edge collaboration but higher computational expense. Li et al. [8] merged AOD-Net and YOLOv5, with robust performance but fine-tuned dehazing. Khan et al. [9] aimed at fast, hybrid models for embedded platforms. Finally, Liu et al. [10] enhanced YOLOX with built-in attention and dehazing, with good performance in moderate fog but poorer performance in dense haze.

A. Research Gap

Despite significant progress achieved in object detection in adverse weather conditions, high accuracy detection in foggy conditions remains difficult. Methods such as Non-Local Image Dehazing, multi-scale object detectors such as YOLOv10, and two-stage architecture such as IDOD-YOLOv7 have proven promising. While models such as ODD-Net and SD-YOLO-AWDNet focus on integrating dehazing and detection, most of them are greatly dependent on synthetic haze datasets or lack in handling extreme conditions such as heavy fog or low-light conditions. Further, most of the methods are computationally intensive or have slow inference times, making them not applicable for real-time applications. This calls for a significant necessity for a unifying and light-weight model combining haze removal and object detection into a single efficient pipeline. This is the problem solved in this work with a proposed light-weight haze-aware detection architecture optimized for real-time usage and trained using a heterogeneous mix of real and synthetic foggy datasets.

III. DATA COLLECTION

A. Overview of the Dataset

The hybrid method, which involves image improvement and object recognition, was tested on two various datasets that exhibit both clear and foggy conditions. The Vehicles Detection Dataset employed the YOLOv10 model to learn from non-foggy images, and it is a clear and well-marked vehicle example source. The dataset contains 1,254 images in total, with 878 for training, 126 for testing, and 250 for validation. **Fig. 1**, provides a visual summary of the training images, showing the diversity and clarity of scenes used to train the model.



Fig. 1. Preview of the Training Dataset

The improvement process greatly enhanced the visibility and distinguishability of objects that were hidden by haze. Thus, the YOLOv10 model performed with higher accuracy in detecting vehicles in scenes that otherwise would be impaired by low visibility. The employment of two datasets enabled a comprehensive evaluation, emulating both ideal and adverse conditions. Lastly, this guarantees that the proposed system performs well not only under ideal conditions but also reliable under adverse environments such as fog, which is essential for application in autonomous driving and traffic monitoring.



Fig. 2. Preview of the Validation Dataset

B. Preprocessing

For the non-foggy training set, a structured series of preprocessing operations was used to maximize the effectiveness and uniformity of YOLOv10 training. These operations involve resizing all images to a uniform resolution, pixel value normalization, and data augmentation operations like horizontal flipping, brightness change, and random scaling. These operations are critical to enhancing generalization over different environmental conditions and object appearances that occur in real-world autonomous driving environments.

First of all, all training images were resized to a constant resolution of 640×640 pixels in order to support the YOLOv10 input requirements. Resizing maintains aspect ratio by proportionally scaling height and width as illustrated in Equation (1):

$$I_{\text{resized}}(x, y) = I_{\text{original}}(x \times W_{\text{orig}}/640, y \times H_{\text{orig}}/640) \quad (1)$$

Here, $I(x, y)$ represents the pixel intensity at coordinates (x, y) , and W_{orig} , H_{orig} are the original width and height of the input image. After resizing, pixel normalization was carried out to normalize the original 8-bit pixel intensity values into the $[0, 1]$ interval. This normalization speeds up training convergence and avoids explosion of gradients in deep layers, as shown in Equation (2):

$$I_{\text{norm}}(x, y) = 255I_{\text{resized}}(x, y) \quad (2)$$

In this equation, $I_{\text{norm}}(x, y)$ is the normalized pixel value, and $I_{\text{resized}}(x, y)$ is the resized image intensity. In order to impose geometric variety, random horizontal flip was used with probability 0.5. It enables the model to learn reversed spatial characteristics and enhances object detection accuracy for entities showing up either on the right or left of the vehicle (Equation (3)):

$$I_{\text{flipped}}(x, y) = I_{\text{norm}}(W - x - 1, y) \quad (3)$$

Here, W is the image width, and the flipping operation mirrors the pixel across the vertical axis. Additionally, the brightness was adjusted by introducing a small random scalar

value to the pixel intensity. This mimics changing lighting

conditions like sunlight, entry into a tunnel, or nighttime driving, as shown in Equation (4):

$$I_{\text{bright}}(x, y) = I_{\text{norm}}(x, y) + \beta \text{ where } \beta \in [-b, +b] \quad (4)$$

In this case, β is a small random brightness offset value within the range $[-b, +b]$, which alters brightness randomly to simulate lighting variability.

Finally, random scaling was employed to mimic varying object distances. Images were scaled by a scale factor s randomly chosen within a specified range, then cropping to preserve the 640×640 resolution (Equation (5)):

$$I_{\text{scaled}}(x, y) = I_{\text{norm}}(s \cdot x, s \cdot y), s \in [0.8, 1.2] \quad (5)$$

Here, s is the random scaling factor used to simulate objects at different distances by enlarging or shrinking them.

The preprocessing operations improved the training set by allowing the model to learn object size, light, and position variation. The validation set within the foggy scene was not processed externally. Fog images were input directly into the hybrid model, where the Non-Local Dehazing (NLD) module operated internally before object detection. The approach supports real-time processing of low-quality inputs without pre-processing the raw data.

C. feature extraction

Feature extraction is carried out in the YOLOv10 architecture, which adopts deep convolutional layers and a light-weight backbone to extract multi-scale spatial and semantic features from images. Better feature representation allows precise object localization and classification, even in cases of partial occlusion or poor visibility. By learning on clean images and testing on hazy images, the model learns robust representations and is evaluated for resistance to atmospheric distortions. The hybrid solution enables more robust feature extraction by combining visibility restoration with detection.

D. Annotation and Label Format

Bounding box annotations for both training and validation sets in the default YOLO annotation format were applied, where an object in one image is assigned a single row of five numbers. The format of the structure of annotations is given as:

$$(\text{classId}, x_{\text{center}}, y_{\text{center}}, \text{width}, \text{height}) \quad (6)$$

The normalized bounding box parameters are computed using:

$$x_{\text{center}} = \frac{x_{\text{min}} + x_{\text{max}}}{2W}, y_{\text{center}} = \frac{y_{\text{min}} + y_{\text{max}}}{2W} \quad (7)$$

$$w = \frac{x_{\text{max}} - x_{\text{min}}}{W}, h = \frac{y_{\text{max}} - y_{\text{min}}}{H} \quad (8)$$

Annotations were created and handled with common tools that support YOLO-compatible formats. This format allows for effective parsing of object location and class data during model training and inference.

IV. PROPOSED METHODOLOGY

To facilitate accurate detection of objects in fog, a deep learning method is proposed that combines detection and image enhancement. For real-time autonomous driving applications, the model initially applies Non-Local Image

Dehazing (NLD) for visibility enhancement and then YOLOv10 for detection. This modular framework combines classical image processing with deep learning for robust performance even under poor weather.

A. Hybrid Model Overview

A hybrid deep learning model integrates Non-Local Image Dehazing (NLD) with YOLOv10 to improve object detection in foggy environments. NLD enhances image clarity by removing haze, while YOLOv10 detects objects efficiently. This unified pipeline boosts accuracy and reliability, making it ideal for autonomous driving in low-visibility conditions.

B. Model Architecture

The architecture is composed of two consecutive modules: NLD Module: Accepts the hazy image as input and produces a visibility-restored image based on non-local patch similarities. YOLOv10 Module: Handles the dehazed image and carries out high-accuracy real-time object detection. The modules are correctly integrated in the pipeline but run independently in their own domains—i.e., object detection and image dehazing. This facilitates plug-and-play architecture that can be easily extended to accommodate future dehazing or detection advancements.

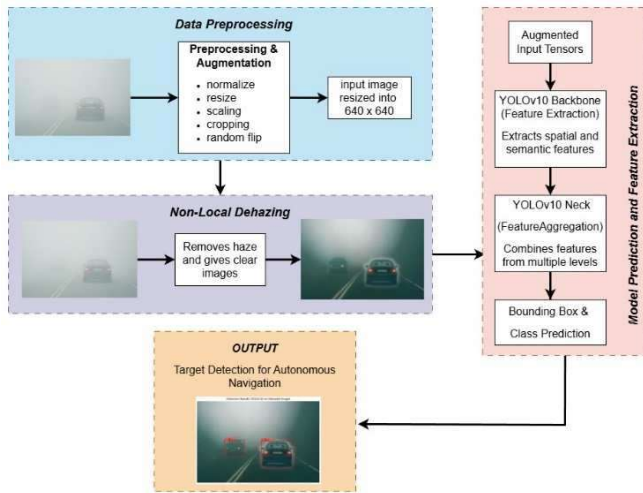


Fig. 3. Hybrid Deep Learning Model Architecture

C. Algorithm 1

1. Input the foggy image $I(x)$
2. Estimate the atmospheric light A using the brightest pixels
3. Estimate the transmission map $t(x)$ based on non-local color line analysis
4. For each pixel x in the image:
 - Calculate the scene radiance using:

$$J(x) = \frac{I(x) - A}{\max(t(x), t_0)} + A \quad (9)$$
 - Clip pixel values to ensure valid range
5. Output the dehazed image $J(x)$

Algorithm 1 aims to enhance the clarity of foggy images by eliminating the haze layer. The input image $I(x)$ represents the original scene affected by fog, where each pixel appears dim due to suspended particles in the air. To recover the clear scene, the algorithm first estimates the atmospheric light A , which accounts for the brightness caused by scattered fog particles. Next, it computes a transmission map $t(x)$, which indicates how much of the scene is visible through the haze at each pixel location. Using these values, the algorithm reconstructs a clearer version of the image, denoted as $J(x)$, using Equation (9). A small constant, t_0 is included in the formula to prevent division errors when the transmission is very low. Finally, pixel values are clipped or scaled to keep them within a valid range, resulting in a visually enhanced, dehazed image. This improved image significantly boosts the performance of the object detection model (YOLOv10) in foggy weather conditions.

Fig. 4. Visual comparison of image dehazing. Left image: Original hazy input, where the visibility is severely impaired by the effect of atmospheric particles. Right image: The efficacy of the NLD (Non-Local Dehazing) algorithm, where the clarity and contrast are improved by eliminating the haze and recovering image details. The process is important for the performance enhancement of vision-based algorithms for autonomous driving and surveillance.



Fig. 4. The original image and dehazed image

D. Algorithm 2

Algorithm 2 explains the process of detecting objects from the dehazed image using the YOLOv10 model. Once the Non-Local Image Dehazing (NLD) module enhances the image visibility, the resulting clear image $J(x)$ is passed into YOLOv10 for analysis. The algorithm begins by initializing the model with pre-trained weights and then processes each image frame to identify object locations, confidence scores, and class labels. During training, the model optimizes a combined loss function that includes bounding box regression, objectness confidence, and classification accuracy. These components guide the model in learning how to correctly identify and classify objects in varying conditions. After prediction, Non-Maximum Suppression (NMS) is applied to filter overlapping bounding boxes and retain the most confident detections. The algorithm ultimately outputs an image with labeled objects, enabling the system to make informed decisions, especially in foggy environments where visibility is normally compromised. This

approach significantly enhances the performance of object detection systems under low-visibility scenarios.

V. RESULTS AND EVALUATION

The proposed hybrid model was tested on both fog-free and fog datasets to confirm its effectiveness in practical autonomous driving situations. Utilizing Non-Local Image Dehazing (NLD) significantly enhanced visual transparency, enabling enhanced performance of the YOLOv10 detector in low-visibility situations. The testing utilized common object detection metrics to calculate classification accuracy, localization accuracy, and general detection credibility.

A. Quantitative Evaluation Metrics

The performance of the proposed model was quantitatively assessed using standard evaluation metrics: Accuracy, Precision, Recall, and F1-Score, as defined in Equations (14)–(17).

$$Accuracy = \frac{TP+FP}{TP+TN+FP+FN} \quad (14)$$

$$Precision = \frac{TP}{TP+FP} \quad (15)$$

$$Recall = \frac{TP}{TP+FN} \quad (16)$$

$$F1 - Score = 2 \times \frac{Precision \cdot Recall}{Precision + Recall} \quad (17)$$

B. Comparison of Models

Algorithm	Evaluation Metrics			
	Precision	Recall	F1-score	Accuracy
YOLOv5 + AOD-Net	0.76	0.72	0.74	0.73
YOLOv7 + IDOD	0.82	0.79	0.80	0.81
SD-YOLO-AWDNet	0.85	0.83	0.84	0.83
YOLO-Vehicle-Pro	0.87	0.85	0.86	0.85
Proposed Hybrid (NLD + YOLOv10)	0.91	0.89	0.90	0.90

TABLE 1 OBJECT DETECTION PERFORMANCE IN FOGGY CONDITIONS

E. Impact of Dehazing (NLD) on Detection Performance

The Non-Local Image Dehazing (NLD) module's effect on object detection was evaluated by comparing the YOLOv10 model with and without the NLD pre-processing step. The addition of NLD greatly improves detection performance in foggy environments, increasing precision, recall, and overall accuracy, as Table 2 illustrates. This illustrates the significance of visibility restoration as a pre-processing step in low-visibility autonomous navigation.

Model	Pecision	Recall	F1 Score	Accuracy
YOLOv10 (Without NLD)	0.78	0.75	0.76	0.77
Proposed (NLD + YOLOv10)	0.91	0.89	0.90	0.90

TABLE 2 YOLOV10 PERFORMANCE WITH VS. WITHOUT NLD

C. Results

Figure 5 plots the train and validation measures of the model built over 100 epochs. Sustained decline in Train Box Loss, DFL Loss, and Class Loss indicates optimal error minimization. Measures such as Precision, Recall, mAP50, and mAP50-95 indicate sustained improvement, especially at initial epochs, indicating improved detection performance. Especially notable is how mAP50-95 saturates after roughly 30 epochs, indicating peak detection across IoU thresholds. On the validation side, losses decrease, and Validation Class Loss sharply declines in the beginning before plateauing, indicating good generalization.

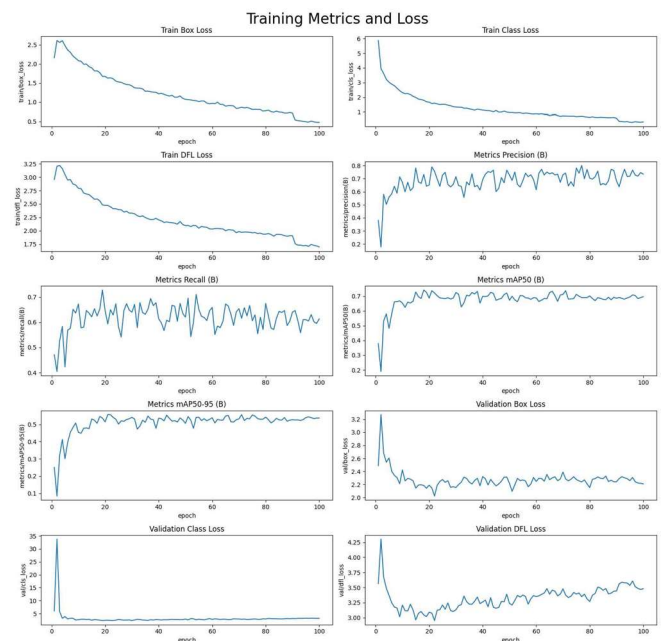


Fig .5 Training and Validation Metrics and Loss Curves

F. Input and Output Results

The above figure illustrates the evolution of an input fog image to a dehazed image through NLD.YOLOv10 model then goes on to process this enhanced image to detect objects. The findings indicate greater visibility and correct identification under low-visibility.



Fig. 6. The original foggy image



Fig. 7. The Output image

VI. CONCLUSION AND FUTURE WORK

A hybrid deep learning architecture is proposed through the integration of Non-Local Image Dehazing (NLD) and YOLOv10 to improve object detection in foggy scenes, resolving significant issues associated with autonomous navigation. The integration of dehazing into the base model

structure improves scene perception and object localization in low-visibility environments. The proposed hybrid technique improves perception reliability, thereby providing safer and more reliable autonomous driving. Model optimization for real-time system application, expanding its applicability to other adverse weather conditions, and its integration with embedded vehicle platforms for end-to-end autonomous navigation are part of the future work.

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